



Examination Report

Biology

6884

EGCSE

2025

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**Paper 6884/01
Short Answer**

General Comments

This is a short answer question paper testing about 80% knowledge and understanding of biology concepts, while 20% concepts require handling of given information. It is marked out of 40.

The 2025 paper was well balanced as questions came from all sections of the syllabus. The paper was accessible compared to that of 2024. As a result, the performance in 2025 seemed to be better than that of the previous year. There was also a decrease in the number of candidates who left blank spaces, and who got zeros.

However, most candidates still failed to bring out the desired responses, thus getting lower marks than what was expected. Factors such as incorrect spellings, insufficient use of biology concepts, and language barrier and failure to read and understand the question, led to candidates failing to score marks. The highest score obtained was 39 while the lowest was 0.

Comments on specific questions

Question 1

This question was accessible to a majority of candidates. The question required candidates complete the hierarchical order in the levels of classification of living organisms. However, some candidates gave incorrect spellings for the phylum such as Phlym, Phylum, Phyleum, Filam, Filum to mention a few.

Expected response: *phylum + order*

Question 2

This question was accessible to most candidates. The question was based on 2 features common in both insects and arachnids. However, some candidates gave characteristics of either an insect or an arachnid as their responses, e.g., 3 pairs of jointed legs, 2 body segments and they lost the mark.

Expected response: *segmented bodies; jointed legs; exoskeleton*

Question 3

Candidates were given a table to complete based on the type of tooth, feature and functions of the teeth. Most candidates lost marks due because they gave an incorrect response of canine instead of incisor. Some gave incorrect spelling such as insizer or insisor and an incorrect function of tearing and chewing. Some candidates failed to give cusps as a response but gave cups, caps or crown as their responses

Expected response: *incisor + cutting*

*3 or 4 roots / presence of cusps/grooved surface + grinding / crushing /
chewing;*

Question 4

This question assessed students understanding of the heart.

- (a) Candidates were expected to give one possible cause of a blockage of a coronary artery. It was accessible to most candidates with a few giving just fats or unsaturated fats of fatty acids, blood clots which led them to lose marks.

Expected response: *plaque buildup / high fats / cholesterol / blood clots / smoking / stress / genes / age / obesity / high blood pressure / lack of exercise / fatty deposits*

- (b) This question was inaccessible to a lot of candidates. The question required candidates to state a right atrium as label A. Most gave incorrect responses such as vena cava, aorta, pulmonary artery and right ventricle.

Expected response: *right atrium*

Question 5

Most candidates could not correctly interpret this question. Candidates were expected to give symbol equations for products of photosynthesis. Instead of writing symbols, most gave the whole equation, while some used words only or words with symbols, and lost the marks.

Expected response: *$C_6H_{12}O_6$ and O_2*

Question 6

This was generally well done. Candidates were able to choose words from the box to complete a statement on emphysema.

Expected response: *smaller + diffusion*

Question 7

This was a fairly attempted question. Candidates were given a diagram of a seedling in a box with a source of light from one side of the box. They were expected to state the response shown by the seedling. A few candidates gave phototropism only without the positive and they lost the mark.

Expected response: *positive phototropism*

Question 8

- (a) This was an inaccessible question. A majority of candidates failed to define an ecosystem and lost a mark, but instead defined a community.

Expected response: *community of interdependent organisms + their environment*

- (b) This was a fairly attempted question. A few candidates came up with their own food chain using different organisms such as mouse, and fox. Some gave arrows pointing in the opposite direction of the energy flow and this resulted in the loss of marks.

Expected response: *(correct order) grass cow ticks birds ;*

three arrows → showing energy flow correctly

i.e. from grass to cow + from cow to ticks + from ticks to birds ;

Question 9

This question was accessible to candidates, though most scored 1 mark. Candidates were expected to match boxes associated with body defences against pathogens. Some candidates matched two responses to one defence and lost a mark.

Expected response: *active immunity - by vaccination*

passive immunity – by antibodies taken from another person

lymphatic system - by producing lymphocytes

phagocytosis – by ingestion and digestion of bacteria;

Question 10

This was an accessible question to most candidates. They were expected to describe a visible structural difference between an artery and a vein. However, some had no idea about a lumen so gave responses using an opening, a hole and space. Other common incorrect responses were a thick cell wall, presence of vacuole or nucleus.

Expected response: *artery is thick walled + vein is thin walled;*

artery has a small lumen + vein has a wide lumen;

Question 11

This question was based on a test and a positive result for Vitamin C. It was accessible to most candidates. Candidates who failed to score marks misspelt DCPIP e.g., DCPD or DCIP, while some gave a starch test as their response. Some candidates failed to give the correct positive test and lost a mark. Their common wrong responses included transparent or clear or colour disappears or fades instead of stating the colour change.

Expected response : *test - DCPIP*

positive result - blue to colourless

Question 12

Candidates were expected to state the function of the medulla. However most failed to score a mark because they gave general functions of the brain. Common wrong response included for memory, for thinking, for involuntary actions, for temperature regulation.

Expected response: *controls or regulates breathing rate / heartbeat / involuntary actions;*

Question 13

This question was fairly attempted by candidates. It showed organs associated with regulation of glucose in the blood.

- (a) This question was accessible to most candidates. Candidates were expected to name organ B as the liver. However some lost marks due to incorrect spellings such as lever, leaver, and liva.

Expected response: *Liver*

- (b) This question was inaccessible to a lot of candidates. Most failed to comprehend that the question required a name of condition but gave description of diabetes. The most common wrong responses was high sugar

Expected response: *diabetes*

Question 14

This was challenging to most candidates. Candidates were expected to list body organs where meiosis occurs. Most candidates gave wrong organs as their responses, while some gave the sperm and egg wrong responses. Other common wrong responses include sex organs, gametes, heart, liver, kidneys

Expected response: *ovaries and testes*

Question 15

This was a fairly attempted question. It required candidates to state why bacteria are useful in genetic engineering. However some candidates gave advantages of asexual reproduction and lost the marks. Common incorrect responses such as bacteria are inexpensive, have a nucleus, they multiply easily, bacteria are unicellular, and produce fast. Common misconceptions were that candidates wrote 'produce' instead of 'reproduce', while some wrote 'easily' instead of 'fast', so lost a mark.

Expected response: *reproduce fast / presence of plasmids / DNA not enclosed / large numbers of bacteria can be grown in a small space / process is inexpensive / no ethical concerns/ no allergies / has restriction enzymes*

Question 16

This question on a chromosome and bases of a DNA was fairly attempted.

- (a) This question was challenging to most candidates as they failed to define a chromosome.

Expected response: *thread of DNA + made up of genes;*

- (b) Generally well done. Some candidates used sex chromosomes, X and Y, in their pairing and no mark was awarded. Some used alleles, e.g. Tt or Bb, instead of bases, and lost a mark. Some remembered the bases but failed to pair them correctly

Expected response: *A and T; C and G;*

Question 17

Candidates were provided with two drawings of eyes in bright light and in dim light and asked to explain how a change in size of a pupil occurs in bright light. Most candidates were able to refer to the diagrams to describe that the pupil decreases in size. However, most candidates failed to score the second mark on the explanation using the muscles associated with pupil reflex. Some confused circular muscles with ciliary muscles and lost the mark

Expected response: *explanation - circular muscles contract + radial muscles relax;
change - pupil become small(er);*

Question 18

- (a) This was an accessible question to most candidates as they were able to use provided arrows to state one for cross pollination. Some candidates gave J as their response, which resulted in the loss of a mark.

Expected response: *H*

- (b) Candidates were expected to state the agent of pollination for a flower they were given. A majority of candidates gave wind as their response and lost a mark. Some gave examples of insects such as bees and lost a mark. Another common wrong response was insect pollination.

Expected response: *insects*

Question 19

This was fairly accessible to candidates. Candidates were expected to state two effects on the human body for long term alcohol abuse. Some candidates gave social and personal effects, and short term effects as their responses and lost the marks.

Expected response: *liver cirrhosis; obesity; brain damage / other organs damage*

Question 20

This was general question on HIV transmission. It was accessible to almost all candidates.

- (a) A few failed to state a precaution but gave only gloves as their response which had no mark. Common incorrect spellings include: glaves, glavs, gloves, and glabs.

Expected response: *wear of safety gloves*

- (b) A few candidates failed to express themselves but wrote 'use protection' which could not score a mark

Expected response: *ref. to male circumcision; ref. to PMTCT; use of condom; ref. to ARV's/ early testing AVP*

Question 21

This question was based on a developing fetus in the uterus. It was fairly attempted by candidates though most failed to score both marks.

- (a) Some candidates lost the mark because they gave responses for the functions of a placenta and amniotic sac.

Expected response: *cushions / act as shock absorber / ref. temperature regulation/
lubricates vagina during birth/ allows movement of fetus / helps fetal
lung development / protects fetus from physical damage*

- (b) Candidates could not state the correct waste product but gave wrong responses such as deoxygenated blood, oxygen, urine and nutrients.

Expected response: *urea / carbon dioxide/ uric acid / nitrogenous waste*

**Paper 6884/02
Structured Theory**

General Comments

There were 15979 candidates who wrote this paper. This means that there was no significant of candidates since there were only 15963 candidates in 2024.

There was a decline in performance compared to last year. Those who did well scored between 45 and 55. There were very few candidates who scored above 55 and the highest mark was 67. There was an increase in the number of candidates who got an overall mark of 0 and many candidates obtained less than 10 of the 80 marks.

Questions that seemed easily accessible to candidates were **1(a)**, **2(a)**, **2(d)(ii)**, **6(a)** and **7(d)(ii)**. The most challenging questions were **1(d)**, **2(c)**, **3(a)(ii)**, **4**, **6(d)(i)**, **7(c)**, **7(e)**, **8(a)** and **8(c)**.

Most candidates were not able to express themselves, showing that the English language is still a big challenge. This led to loss of marks. For example,

- (a)** Question **1(c)** candidates would write “yeast produces carbon dioxide” instead of “enzymes in yeast convert sugar into carbon dioxide”,
- (b)** Question **2(c)** candidates mentioned the liver as doing the emulsification instead of bile salts which are produced by the liver.
- (c)** Question **4(b)** candidates would write “the relay neurone tells the motor neurone” instead of “the impulse is sent from the spinal cord to motor neurone via the relay neurone”.

Comments on individual questions

Question 1

This question proved to be challenging to most learners. Most candidates were able to score 1 out of 8, mostly from **(a)**.

- (a)** This question was well done. Candidates were able to state the type of reproduction as *asexual reproduction*. However, some candidates lost marks by adding binary fission thus making the response unacceptable. Other common wrong answers included unsexual reproduction and sexual reproduction.
- (b)** The question was challenging for most candidates. A majority of candidates failed to state the function of the ribosome. Some wrong responses included site for respiration, for metabolic reactions, protection for the nucleus and for photosynthesis.

Expected response: *site for protein synthesis*

- (c)** This question was challenging to most students. A number of candidates were describing anaerobic respiration in muscles, stating the products of anaerobic respiration in yeast as

lactic acid and carbon dioxide. Most had an idea that the dough rises, however, they thought it was due to the yeast instead of carbon dioxide or that the yeast has bacteria which makes the dough rise. Some candidates were just outlining the method of bread making instead of describing the role of yeast in bread making.

Expected response: *enzymes in yeast breakdown sugar and release carbon dioxide and alcohol. Carbon dioxide makes the dough rise*

- (d) This question proved to be more challenging to most candidates. A few candidates were able to score 1 out of 3 points. They were required to describe how fungi are adapted to their environment. Most candidates stated the features of fungi without properly stating the advantage of having the feature.

Expected response: *many spores produced + increased chance of reproduction
light/small spores + for dispersal by air currents / to withstand harsh conditions
mycelia/ hyphae + large surface area for absorption of food materials
sporangium held away from the substrate + better spore dispersal
sporangium bursts + to assist in spore dispersal/ to expel spores into the air*

Question 2

This question was fairly done by most candidates.

- (a) This question was well done, as most candidates were able to name the *oesophagus/ gullet*. Some were giving wrong spellings such as oesophegus, osofogas, etc. Some wrongly identified the oesophagus as the trachea, duodenum, throat or alimentary canal.
- (b) The majority of the candidates found this question challenging. They were required to describe the digestion of fats in the stomach. Most could not obtain the mark for mentioning the acidic pH suitable for enzyme activity. Some candidates lost a mark for the enzyme by stating that it is produced by the pancreas.

Expected response: *pepsin
in an acidic pH/pH 2
proteins digested to peptides/ polypeptides*

- (c) The question was challenging to most candidates. Those who attempted to give good responses lost a mark for writing no digestion of fats instead of slow digestion. Weaker responses described the functions of the liver and how the liver is affected by alcohol or the importance of the liver in digestion instead of the effect of liver damage on fat digestion. There were a lot of misconceptions in most of the answers. Some were stating that bile is

an enzyme that digest fats. Others stated that the liver emulsifies or digests fats. Some candidates thought that liver damage will result in the accumulation of fats leading to coronary heart diseases.

Expected response: *less bile production/ less bile salts*
less emulsification of fats
slow digestion of fats

- (d) (i) The question was accessible to most candidates. They were able to state the features of the ileum that make high levels of absorption possible. Some wrong responses included stating the presence of lacteals, mentioning blood capillaries without qualifying them or stating villus instead of villi.

Expected responses: *it is long*
has thin epithelium/ one cell thick
numerous villi
has a dense network of blood capillaries

- (ii) This was a challenging question for some candidates. They were asked to state the route taken by glucose from the time it is absorbed by the ileum until it reaches the kidneys. Some confused the pulmonary artery for the pulmonary vein. Most candidates failed to obtain a mark for **F**, where they were expected to name two blood vessels for 1 mark. Most of those who attempted to give correct responses only named one blood vessel and they lost the mark.

Expected responses: **B** - hepatic portal vein
C - hepatic vein + vena cava
D - pulmonary artery
E - pulmonary vein
F - aorta + renal artery

Question 3

This question was challenging to most candidates.

- (a) (i) The question was well done. They were given a leafy shoot immersed in a red dye solution. Candidates were able to label the *xylem* in the given cross - section of a stem as the area where the red dye absorbed by the plant will be seen. However, some left the label line without the name and some labelled it as phloem or pith. Some candidates labelled Fig. 3.1, which was showing the shoot put in the red dye solution, instead of Fig. 3.2 which was the cross section of the stem.
- (ii) The question proved to be challenging to a majority of candidates. Candidates were expected to explain how the red dye solution had reached the veins in the leaves.

Most candidates stated that the red dye solution moved up the plant using incorrect processes such as osmosis, diffusion, active transport and translocation. Some assumed that the solution was moving through the phloem. There was also incorrect use of cohesion and adhesion in the explanations.

Expected responses: *transpiration pull*

cohesion

adhesion forces

water/ dye moved up into leaves

- (iii) The question was fairly done as most candidates were able to score 1 out of 2 marks. They were able to describe that the leaf will wilt if taken out of the solution. However, they could not explain the wilting. They were explaining in terms of the leaf cells being flaccid, plasmolysed and shrinking thus losing marks. Some stated that the plant will lose water without mentioning the process of transpiration. Those who mentioned transpiration thought there will be more transpiration than water absorption yet the plant was already taken out of the solution.

Expected responses: *wilting*

water lost by leaves through transpiration has not been replaced

- (b) The question proved to be challenging to candidates. They were expected to state differences between the cross- section of a root and that of a stem. Most were writing incomplete answers such as xylem at the centre or xylem star shaped, without referring to the phloem.

Expected answers: *no pith*

vascular bundle at the centre/ xylem star-shaped and

phloem at the arms of the star

no vascular cambium

presence of root hairs

Question 4

This whole question was very challenging to most candidates.

- (a) Candidates were expected to describe and explain how structure **G** (arteriole) and **H** (sweat gland) help to keep body temperature constant when external temperature is 5 °C. Some candidates referred to **G** as hairs/ capillaries that constrict and lost marks. Some made errors of stating that **G** contracts instead of constrict. They also lost marks when using terms such as prevent/ stop/ avoid heat loss instead of reduced/ less heat loss. Some

candidates referred to **H** as an erector muscle instead of sweat gland. Some also had a misconception that sweat pores close in order for sweat not to be released instead of stating that sweat glands do not produce sweat.

Expected answers: (G) constrict/ narrow
less blood flow near surface
(H) sweat production stops / reduced
ref to less heat loss in G/H
ref to radiation in G

- (b) This question was difficult for most candidates. Most candidates seemed not to have an idea of a reflex arc. They could not describe a reflex arc that would cause a person to remove his hand from a hot object. They failed to make reference to transmission of electrical impulses from sensory neurone to central nervous system, from relay neurone to motor neurone and from motor neurone to effector. Those who did could not properly show the pathway of the electrical impulse. A few referred correctly to the synapse. Some responses that resulted in loss of marks included failing to make reference to synapses, stating that sensory neurones transmit nerve impulses to the brain instead of spinal cord and stating the sensory neurone as a receptor.

Expected responses: sensory neurone to spinal cord
relay neurone to motor neurone
ref to synapses
motor neurone to effector muscles

Question 5

The question was fairly done. Most candidates obtained 4 out of 8 points. The question was about an investigation to find out the effect of light intensity on the rate of photosynthesis. A lamp was used a source of light and was put at different distances from the water plant. The number of bubbles produced were counted to compare the rate of photosynthesis per distance. The results of the investigation were shown in a table.

- (a) The question required them to suggest and explain why the water plant was kept at an optimum temperature. Most candidates stated that temperature should not be a limiting factor instead of stating that temperature is a factor that affects the rate of photosynthesis. Others thought that it will make the investigation to work well. They also failed to make reference to denaturing of enzymes by high temperatures. There was also no reference to diffusion rate of carbon dioxide or kinetic energy.

Expected answers: temperature is a factor that affects the rate of photosynthesis
photosynthesis is controlled by enzymes that act at an optimum

temperature

too high temperatures denature enzymes

idea of diffusion rate of carbon dioxide

reference to kinetic energy

- (b) The question was fairly done. Some were able to describe and explain the effect of moving the lamp from 10 to 20cm from the water plant. Those who lost marks include those who were transferring the data from the table without giving an explanation and interpreting the given information.

Expected responses: *more bubbles released at 10 cm than at 20 cm*

higher light intensity at 10 cm than at 20 cm

faster rate of photosynthesis at 10 cm than at 20 cm

light a limiting factor at 20 cm

- (c) The question was fairly done. Some candidates were able to explain why seedlings failed to grow in a soil lacking magnesium when all conditions for growth are available. Some candidates gave incomplete answers such no magnesium thus no photosynthesis, there was lack of magnesium, there was lack of chlorophyll synthesis resulting to no growth.

Expected responses: *magnesium needed for chlorophyll formation*

no chlorophyll + no photosynthesis + no growth

Question 6

This question was fairly done. A majority of the candidates scored 5 out of 10 or more.

- (a) This question was well answered. Most candidates were able to state the process taking place at J. A few candidates described the process instead and lost a mark.

Expected answer: *ovulation*

- (b) This question was poorly done. Candidates were required to describe and explain changes in the uterus lining during the menstrual cycle from Fig. 6.1. For the description, most candidates were giving responses such as “it grows, it builds up, it repairs, it enlarges, it becomes wide” instead of “it increases”. Some lost marks for writing “thicker uterus” instead of “thicker uterus lining”. The candidates were expected to explain that the increase in the thickness of the lining was for readiness for implantation if fertilisation takes place. They were also expected to state that the cause for the increase in thickness is the rise in the level of oestrogen. However, some described implantation incorrectly. For example, they said it was for the attachment of the ovum or even ovule. Some of them failed to mention the rise in oestrogen level. Some mentioned progesterone which showed failure to refer to the given figure.

Expected responses: (description): *thickness increases*

(explanation): (cause) *because oestrogen level rises*

(reason) *to prepare for implantation of fertilised egg*

- (c) This question was poorly done. Most candidates failed to state that oestrogen reaches its target organs by being *transported in the blood/ plasma*.
- (d) (i) The question was fairly done. Some candidates were able to describe and explain ways in which hormones in fertility drugs treat infertility in females. Some were able to name the hormones but mixed up the functions. Some were naming both hormones but giving only one function, suggesting that both hormones carry out the same function. Other wrong responses included naming wrong hormones or including wrong hormones to the correct ones.

Expected responses: *use of fertility drugs that contain FSH for development of ova*
LH for ovulation

- (ii) The question was fairly done. Candidates who had challenges mentioned natural methods instead of mechanical methods of birth control. Some wrote wrong responses like contraceptives, abortion or protection.

Expected answers: *use condom / femidom / diaphragm / IUDs*

Question 7

Generally, the question was fairly done. A majority of candidates obtained 6 out of 13 marks.

- (a) This question was fairly done. Most common wrong answers were: natural variation, genetic variation and discontinuous variation. Those who gave the correct response were able to state that the height of seedlings show *continuous variation*.
- (b) The question was fairly done. They were required to state causes of continuous variation in the height of the seedlings. Some candidates were listing examples of environmental factors without qualifying them. For example, “nutrients, light, water” instead of “competition for nutrients, light or water”. They also gave incorrect responses like climate, seed type and soil fertility.

Expected responses: *environment or named such as uneven distribution of nutrients /*
water/ light
genes

- (c) This question proved to be difficult for most candidates. Most candidates lost marks for not labelling the genetic diagram and incorrect labelling. Others lost marks for crossing the genotypes. Others lost marks for using blood group or sex- linked symbols. Some candidates were not aware of what was expected from the question, as they drew maize cobs and seeds instead of the genetic diagram.

Expected response:

parents phenotype white x white
 parents genotype Hh Hh
 gametes (encircled) H h H h

	H	h
H	HH	Hh
h	Hh	hh

offspring genotype HH Hh Hh hh
 offspring phenotype white white white black

- (d) (i) The question was fairly done. Most candidates were able to define mutation however, some lost marks for leaving out “spontaneous/ sudden” in their response.

Expected response: *sudden or spontaneous change in gene or chromosome*

- (ii) This question was well done. Most candidates were able to state the factors that increase mutation rate. Wrong responses include sunlight, sun-rays.

Expected response: *exposure to chemicals / X-ray radiation*

- (e) This question was poorly done. Most candidates generalised on reproduction of bacteria and failed to make reference to mutation, resistance and passing on of genes. Some gave vague answers referring to genetic engineering in the production of insulin while others described active immunity.

Expected answers: *a mutation has resulted in resistance
 that resistant bacterial cell reproduced
 mutation was inherited by all subsequent offspring*

Question 8

This question proved to be difficult for most candidates. A majority of the candidates failed to get at least half of the marks.

- (a) This was a challenging question for candidates. They were expected to explain how dead animals add nitrates in the soil. The question required them to describe two processes; decomposition and nitrification. Most candidates lost marks for confusing the two processes, for example, they would write that decomposition produces nitrates in the soil instead of nitrogen compounds or ammonia. Most candidates also lost marks for using the wrong names of bacteria for decomposition or nitrification. For example, they would write that dead animals are decomposed by nitrifying bacteria.

Expected answers: *dead animals decomposed*
nitrification / proteins or other nitrogenous compounds broken down
by bacteria/ fungi

- (b) This question was fairly answered. Candidates were able to correctly name the process as denitrification. Common wrong responses include nitrification, nitrogen fixation, lightning and evaporation.
- (c) This question was fairly done. Candidates were expected to describe how nitrates in the soil are obtained by animals. Most candidates failed to mention that nitrates absorbed by plants are used to make proteins in the plants. Some candidates lost marks when they wrote that nitrates are taken up by osmosis or diffusion.

Expected responses: *(nitrates) absorbed by plants to make proteins*
animals feed on plants

- (d) This question was poorly done. The candidates were expected to suggest why extinction of elephants should be prevented. Some wrote wrong answers like the importance of conserving endangered species such as medicinal use. Some stated that elephants represent the queen mother. Only roles targeted to the elephant were accepted.

Expected answers: *reference to their role in the ecosystem/ stated examples*
reference to food chains/ webs being maintained

**Paper 6884/03
Practical Test**

General comments

Biology Paper 3 is a practical examination paper to test Assessment Objective **C** of the Assessment syllabus. It aims at assessing the level of candidate achievement in investigative and manipulative skills embracing the scientific method of inquiry. In its nature the paper demands that candidates be exposed to as much practical activities and the scientific process skills as possible. Candidates need to be familiar with basic laboratory equipment and apparatus as well as the skills to use them correctly. Furthermore candidates need to be trained on the scientific method of inquiry, including improve experiments ensuring reliability and validity of experimental data and drawing conclusions from the data.

The 2025 paper was of the same difficulty as papers from previous years. It consisted of **two** compulsory questions with a maximum possible score of 40.

Very few centres had difficulty in providing the required examination materials. It is worth noting though that some centre supervisors failed to provide the examiners with useful information in their supervisor`s results which was valuable in the marking of candidates works e.g. failure to state the colour of the fruits yet there was a question that required the candidates to state the colour of their specimens. For some centres the samples of the potato strips were swapped. It was also noted that in some centres the lemon was cut longitudinally instead of being cut transversely).

Though the paper was of the same level of difficulty as the previous year`s, the 2025 highest score was 38 and the lowest was 4 and there were no zeros recorded which showed an improvement compared to 2024.

It was noted with great concern that many candidates lack skill of reading-off and recording measurements from a ruler.e.g. When measuring the length of the potato cylinder they would record 55.5 mm instead of 55 mm or 56 mm. Some failed to convert cm to mm. English language usage still remains a challenge for many candidates and failure to do so results in the loss of marks since it then distorts the scientific facts .The use of comparative language is poor e.g. when comparing the texture of the potato cylinders they would use terms like ‘fresh’ instead of smooth, ‘weak’ or ‘loose’ for flexible.

Both questions were accessible but question **1(a)(ii),1(e), 1(g) (i)(iii)** and question **2 (c)(iii)** proved to be difficult for candidates ,while question **1(b)** proved to be easy.

Comments of specific questions

Question 1

This question was on osmosis. The students were provided with potato cylinders placed in solutions with different water potentials. In the investigation two potato cylinders of same length were placed **in** beakers **A** and **B** and left for 24 hours. The candidates were required to measure the lengths of the potato cylinders, feel their texture and explain their observations.

- (a) (i) A majority of candidates correctly measured and recorded the length. Some lost a mark for using wrong units or failure to read-off the measurements correctly from their rulers. Those that lost a mark had wrong units e.g. 54 cm instead of 54 mm or 50.4 mm instead of 54 mm.

Expected response: *A-more than 50 + B-less than 50 and **both** recorded in mm.*

- (ii) This question required candidates to calculate and record the change in length for each potato cylinder. Many candidates lost the second mark since they failed to record the change in length in cylinder B as a negative.

Expected response: *two changes in length correctly calculated and recorded and the change in **B** recorded as a negative.*

- (b) This question required candidates to examine and record the texture of the potato cylinders. A majority of candidates scored these marks; they easily described **A** as *hard / firm / tough* and **B** as *soft / bending / smooth*. There some though that lost mark for describing using terms e.g. 'fresh' for smooth, ' loose' for soft or bending.

- (c) This question required candidates to explain the observed change in length and the texture of potato cylinder **B**. most candidates realised that the change was due to osmosis hence they simple described osmosis without making reference to the potato cylinder or the salt solution, also without stating the direction of movement of the water molecules. In some responses, it was the solution that was moving into or out of the cylinders.

Expected response: *that water moved out of the potato cells/tissue by osmosis
resulting in cells becoming flaccid/loses turgidity/ plasmolysed.
potato cells become smaller / shorter.*

- (d) Candidates were required to state the dependant variable in the investigation. Some candidates lost the mark by stating same length /temperature or volume of solutions.

Expected response: *change in length / final length.*

- (e) This question required candidates to state variables which should have been controlled in the investigation. Most candidates could not obtain all three marks, some stated same concentration of solution which did not score the mark.

Expected response: *temperature of solutions*
volume of solutions
same diameter of cylinders
both cylinders taken from same potato / same species / type / age
of potato.
no skin included

- (f) This question required candidates to state why repeating the experiment would be important. Some common errors were confusing validity and accuracy for reliability.

Expected response: *to identify anomalous results from the wide range of results*
and thus improving the reliability of results.

- (g) (i) This question required candidates to plot on a grid a line graph using data collected when similar potato cylinders were placed in sucrose solutions of different concentrations. Many candidates lost marks due to failure to identify the dependant and independent variables, the grid with negative values confused them. Some did not use the ruler to join the points but used free hand.

Expected response: *sucrose concentration plotted on the x-axis,*
axis fully and correctly labelled with units
linear scale including zeros used + plotting covering over half
of the grid horizontally and vertically
five points correctly plotted
all five points joined with ruled straight lines

- (ii) The question required candidates to state the conclusion for the investigation. Some candidates lost marks by stating that as the sucrose concentration increases the length of the potato cylinder decreases.

Expected response: *as the sucrose concentration increases the change in length*
decreases or as the concentration of sucrose decreases the
change in length increases.

- (iii) Candidates were required to find the concentration of sucrose in which the length of the potato cylinder did not change. Many candidates could not to indicate on their graphs where the reading was taken hence they lost a mark. Some used the calculator to extrapolate instead of using their graph.

Expected response: *correct indication of sucrose concentration from the drawn graph and correct reading with units.*

Question2.

(a) The candidates were provided with halves of two fruits **C** (raw banana) and **D**-ripe lemon.

- (i) The question required candidates to state one visible similarity in fruits **C** and **D**. Though many candidates stated correct similarity, some lost a mark by stating presence of cell wall/shells or leaves for the peels.

Expected response: *both circular and round/both fleshy/both have peels.*

- (ii) This question required candidates to compare fruits **C** and **D** using a provided table. Common errors were stating raw or ripe, sour or sweet citrus or not citrus.

Expected response: *larger vs. smaller*
thinner fruit wall vs. thicker fruit wall
presence of segments vs. absences of segments
presence of seeds vs. absence of seeds

- (b) This questions required candidates to make a large drawing of fruit **C**. Some candidates lost marks by using a compass when making the outline; others shaded their drawings and had unrealistic drawings.

Expected response: *drawing with a larger diameter than that of C*
clear, continuous outline
no shading
fruit wall indicated with a double line and the fruit wall showing
some segments indicated

- (c) (i) Candidates were required to measure length of diameter of fruit **C**.

Expected response: *a recorded diameter of not less than 30 mm but not more than 70 mm*

- (ii) This question required candidates to draw, measure and record the length of a line passing through diameter of drawing of fruit **C**. Most candidates were able to draw the line as required. However, some had over laps, lines with arrow heads which were penalised. The lack of skill of reading from the ruler and recording length resulted in the loss of marks.

- (iii) This question required candidates to calculate the magnification of their drawing of fruit **C**. Common errors were: swapping lengths of diameter of fruit with that of fruit **C**, omitting X/times as units for magnification. Some students did not round off their answers to the stated number of decimal points.

Expected response: recorded diameter of drawing
recorded diameter of **C**;
correctly calculated magnification to 2 decimal places;
expressed with 'X'

- (d) (i) The question required candidates to record observations made after adding iodine solution onto the cut fruits. Common wrong responses were: no change/ no observation for fruit **C** instead of stating that iodine solution remains brown / golden yellow. For fruit **D** most described the observation as blue black or dark blue. Some common errors were describing the colour as blue/ purple.

Expected response: *specimen **C** remains brown and specimen **D** changes to blue-black*

- (ii) The question required candidates to make conclusions from observation made in (d)(i). Common errors were stating that fruit **C** contained vitamin C while fruit **D** did not. Some lost the marks by stating that it was a negative test or positive test without any explanation.

Expected response: *no starch in fruit C + starch present in fruit D*

**Paper 6884/04
Alternative to Practical**

General Comments

The alternative to practical paper assesses a range of practical skills. Candidates preparing for this paper need the same kind of preparation as the practical test paper.

The overall performance of candidates showed a slight improvement compared to the previous year. Candidates attempted all questions and showed that they had adequate time to finish the paper. Most candidates found the questions on measurements less challenging with a few that had challenges with reading the scale on their rulers. A significant number of candidates produced good diagrams with a few that lost marks for shading and omitting some details. Candidates also found the questions on food tests easy.

A number of candidates struggled with questions that required them to identify variables in the investigation. Graphical skills also proved to be challenging for a significant number of candidates as they failed to get all the marks.

Comments on specific questions

Question 1

- (a) (i) This question was generally well answered by a majority of candidates with a few that had challenges with reading the scale in millimetres. The correct length for the potato cylinders was 50 mm. The mark was awarded for a correct measurement and unit.
- (ii) This question was fairly done by a majority of candidates with a few exceptions of candidates who had challenges with reading the scale on their rulers. Another common error was to omit the units.

Expected response: potato **A** = 54 mm
potato **B** = 47 mm

- (iii) Most candidates were able to calculate the change in length for both potato cylinders. A majority lost the second mark because they omitted the minus sign for the potato cylinder that decreased in length.

Expected response: potato **A** 4 mm
potato **B** -3 mm

- (b) This question proved to be a bit challenging for a significant number of candidates. Candidates were expected to suggest the texture of the potato cylinders after they were left for 24 hours in different solutions based on the change in length.

Expected response: potato cylinder **A** - rough/firm/hard
potato cylinder **B** : floppy/soft/smooth etc.

- (c) This question was fairly done by a significant number of candidates. Candidates that were familiar with the investigation on osmosis were able to explain why potato cylinder **B** decreased in length.

Expected response: *The potato had a higher water potential than the solution so water moved out of the potato tissue by osmosis. The length of potato decreased and the cells became flaccid or lost their turgidity.*

- (d) Many candidates found it challenging to identify *change in length* as the dependent variable in this question.
- (e) Some candidates were able to identify variables that were controlled in the investigation. A significant number however gave vague responses that lack details such as “the potato, the solution” which didn’t score any mark.

Expected response: *temperature of solutions*
volume of solutions
same diameter of cylinders
both cylinders taken from same potato / same species / type / age of potato.
no skin included in cylinder

- (f) Most candidates struggled to state the importance of repeating the investigation. Repeating an investigation assist in *identification of anomalous results from the wide range of results and thus improving the reliability of results*. Candidates should be reminded that repeating investigations does not make the procedure accurate.

- (g) (i) The quality of graphs varied amongst candidates, some were able to plot excellent graphs whilst others struggled to select a suitable scale to plot the given values.

Common errors included:

- omitting labels on axes and writing correct units
- scale that covered less than half of the grid provided
- use of non-linear scale
- incorrect orientation of the axes

Candidates should be reminded that the labels and units of the axes should match the headings and units in the table. Candidates should not extend the line beyond the plotted points unless there is an instruction to do so.

Expected response: *sucrose concentration plotted on the x-axis,
axis fully and correctly labelled with units
linear scale including zeros used + plotting covering over half
of the grid horizontally and vertically
five points correctly plotted
all five points joined with ruled straight lines*

- (ii) A number of candidates were unable to state a proper conclusion but described the trend shown in the table instead. Candidates had to the aim of the investigation when considering their conclusion.

Expected response: *as the concentration of sucrose increased, the potato cylinders decreased in length.*

- (iii) Candidates were expected to use their graphs to find the concentration of sucrose in which the length of the potato cylinder did not change. Candidates that did not show how they obtained their answer were penalised. *Candidates were expected to identify the value on the x-axis where the plotted line intercepts the 0 on the y-axis, draw a vertical line on that point and then read the corresponding concentration.*

Question 2

- (a) (i) Most candidates were able to state a visible similarity between the two fruits. A common error was to mistake the fruits for cells. Another common mistake was to state similarities that were not visible.

Expected response: *both have a fruit wall
both are circular in shape
both have seeds*

- (ii) A significant number of candidates were unable to score full marks in this question. Candidates are to be reminded that they should compare one feature at a time to score a mark. Differences that were acceptable include the following:

Expected responses: *fruit C is large in diameter whilst fruit D is small in diameter
fruit C has a thin fruit wall whilst fruit D has a thick fruit wall
fruit C has seeds/ larger seeds whilst fruit D has no seeds / smaller seeds
fruit C has segments / cross walls whilst fruit D does not have segments
/ cross walls*

- (b) Most candidates were able to make a large drawing of fruit C. Candidates should be encouraged to draw with a sharp pencil to show a clear continuous outline. Shaded drawings are not allowed. All details given in the specimen should be included in the drawing and in the correct proportions.

Expected response: *large drawing with a diameter of ≥ 60 mm*
clear continuous outline
no shading
fruit wall indicated with a double line and the fruit wall showing
some segments
6 seeds drawn

- (c) (i) Most candidates were able to measure the diameter of fruit **C** as 45 mm. Some candidates struggled to read the scale in mm and instead of 45 mm they wrote 40.5 mm. Some candidates wrote down a measurement in centimetres yet millimetres being given.
- (ii) Most candidates were able to draw a line across their drawings and measured the length of their line accurately. Candidates that did not draw the line across the diameter of their drawings as instructed were penalised.
- (iii) Most candidates used their measurements correctly to work out the magnification. Common errors included: dividing the measurement of the photograph (**C**) by the drawing, including units such as mm in the final answer and failing to express the magnification to two decimal places as instructed. Candidates are to be reminded that magnification does not have units and should have the times (X) sign.

Expected response: recorded diameter of drawing
recorded diameter of **C**;
correctly calculated magnification to 2 decimal places;
expressed with 'X'

- (d) This question was well answered by a majority of candidates.

Expected response: *specimen **C** does not contain starch whilst specimen **D** contains starch.*

- (e) Generally well attempted by a number of candidates.

Expected response: *reagent: Benedict's solution*
and that the orange precipitate showed the presence of reducing
sugars.

